Bags, Vectors & Transformers

Ethics, Access & Distillation · Day 4, Part 2

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Whose writing did it learn from? Did they agree?

Labor

Who labeled and filtered for it, and for what pay?

Energy

How much electricity per label?

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Who could have run this, and in which language?

Power

Whose model, whose terms? Can they take it away?

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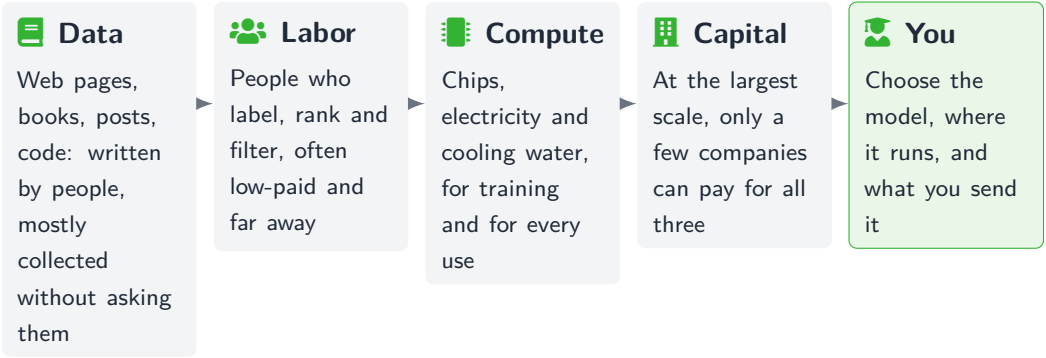
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None of this started with LLMs: every method in this course raises some of these questions, and LLMs make each one bigger. **Distillation**, the last idea of the course, answers several at once.

Behind Every Model: A Supply Chain



A dictionary has a supply chain too, just a much shorter one: someone wrote it, for some purpose, based on some texts. The questions stay the same; the scale doesn't.

Data: Whose Text, Whose Consent?

Large models learn from the crawled web, books, Wikipedia, code and social media. The people who wrote it were rarely asked or paid.

🏛 Closer to home: Cambridge Analytica (2018)

A university researcher's quiz app collected data on about 270,000 users **and their Facebook friends**: up to 87 million profiles, later used for political ads. Collected for research, reused for something nobody agreed to.

🔨 Bartz v. Anthropic (2025–26)

Training on books it had bought was fair use, the judge found; keeping millions of pirated copies was not. Settled for $1.5 billion, about $3,000 per book.

🔨 Kadrey v. Meta (2025)

Meta torrented books from shadow libraries such as LibGen. It won on fair use, but the judge stressed this did not make its copying lawful: the authors had argued the wrong points.

📷 Clearview AI (2024)

Scraped over 30 billion face photos to sell facial recognition. Fined €30.5 million by the Dutch Data Protection Authority.

Labor: The People Inside the Model

Rounds 2 and 3 of training (Part 1) run on people: they write example answers and compare outputs. Others label harmful content for safety filters.

TIME, 2023

Kenyan workers hired through the contractor Sama read and labeled graphic descriptions of violence and abuse for OpenAI's safety filter. Take-home pay: about $1.32–2 an hour. Workers described lasting psychological harm.

In our own research

Crowd workers on MTurk or Prolific code our texts and fill in our surveys. What we pay them, and what we ask them to read, is a research-ethics decision.

A twist for annotation

An estimated 33–46% of crowd workers used an LLM for a summarizing task they were paid to do themselves (Veselovsky et al., 2023). Your "human" labels may not all be human.

Energy: The Bill Comes With Every Use

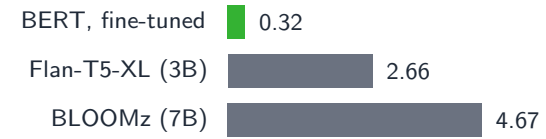
Training happens once. BLOOM (176 billion parameters) emitted about 25 tonnes of CO_2-equivalent, or 50 counting the hardware and idle time (Luccioni et al., 2023).

Use happens millions of times a day. For a popular model, that can add up to more than the training.

Data centers also need water for cooling. Few companies report either in a form others can check.

Sentiment classification, 1,000 texts

grams of CO_2-equivalent



The two general-purpose models were prompted, with no fine-tuning. Luccioni, Jernite & Strubell (2024): such models were consistently more expensive than task-specific ones doing the same task.

Measure your own runs: the Python package `codecarbon` estimates energy use and emissions.

Bias at Every Level of This Course

Bags Day 1

A dictionary is someone's view of language. General sentiment lists call *tax*, *cost* and *liability* negative, which misreads financial reports.

Loughran & McDonald (2011)

Vectors Day 2

Embeddings learn our stereotypes: *man* is to *programmer* as *woman* is to *homemaker*. They hold the same associations as implicit-bias tests.

Bolukbasi et al. (2016); Caliskan et al. (2017)

Transformers Day 3

In hate-speech data, tweets in African American English were up to twice as likely to be labeled offensive, and classifiers learned it.

Sap et al. (2019)

LLMs Day 4

Asked directly, models describe Black people positively. Given text in African American English, they give the writer worse jobs and convict more often.

Hofmann et al. (2024)

Bias sits in the data and the labels, so it travels with them, all the way into a distilled student. **Check the errors separately for the groups you study.**

When a Classifier Decides About People

The childcare benefits scandal (*toeslagenaffaire*)

- The Tax and Customs Administration picked childcare benefit claims for fraud checks with a **risk classification model**. Having Dutch nationality or not was one of its inputs.
- Tens of thousands of parents were wrongly treated as fraudsters and told to pay back benefits, disproportionately families with a migration background.
- The cabinet resigned in January 2021. The Dutch Data Protection Authority fined the Tax Administration €2.75 million for "unlawful, discriminatory" processing of nationality data.

Why this belongs in a text-analysis course

No LLM, no deep learning: a simple model, a biased input, and output used as a decision rather than a measurement.

Your classifiers are measurement instruments. Once their labels start deciding things about people, their errors become harms.

Amnesty International (2021), *Xenophobic Machines*; Autoriteit Persoonsgegevens (2021).

Who Gets to Build, Who Gets to Use

Money and compute

About 70% of new AI PhDs go to industry, up from about 20% in 2004, and industry built the year's largest model 96% of the time (Ahmed et al., 2023). Academics increasingly study models they could never build.

Language

The same content can take up to 15 times as many tokens in one language as in another (Petrov et al., 2023): it costs more, runs slower, and less fits in the context window. Models also tend to work worse in languages with less training data.

Data access

In 2023, Twitter/X and Reddit closed or priced out the free API access that much social science relied on. Since late 2025, the EU's Digital Services Act (Art. 40) gives vetted researchers a route to data from the largest platforms.

In this room

Free or paid Colab, an API billed in dollars to a credit card, a laptop that can or can't run a model. Who in your field could not have done today's notebook?

“Open” Is a Spectrum

“Open” gets used loosely. Ask what exactly is open:

	Weights	Training data	Training code	License
GPT, Claude, Gemini	✗	✗	✗	API terms only
Llama 3.1 (Meta)	✓	✗	✗	custom, with conditions
Qwen2.5 1.5B (notebook)	✓	✗	✗	Apache 2.0
OLMo 2 (Allen AI)	✓	✓	✓	Apache 2.0

Liesenfeld & Dingemanse (2024) rated over 40 “open” generative models and found many big-company models **open-washed**: weights released, training data and methods kept closed. For research, open weights already give you privacy and a version that can’t change under you. Open data also lets you check what the model saw, for instance whether your test set was in it.

The Companies Behind the Models

A pattern	A documented example
Take first, ask later	Meta and Anthropic trained on pirated books (court records, 2025); Clearview scraped over 30 billion faces.
Pass on the costs	Safety labeling for OpenAI at under $2 an hour (2023); energy and water use that few companies disclose in checkable form.
Pull the rug	OpenAI shut down its Codex models in 2023 with under a week's notice; about 100 published papers could no longer be reproduced.
Ship first, fix later	January 2026: Grok on X made sexualized images of real women and children on request. X restricted it after a public outcry; UK and EU regulators opened investigations.
Concentrate power	A few firms control the leading models, the chips they need and the cloud they run on.

Not every company, not all the time. But when you pick a model, you also pick whom you depend on, and whose practices your research helps make look normal.

Your Position as a Researcher

The rules you work under

GDPR. Personal data sent to a commercial API needs a legal basis and a processing agreement. For health data and other special categories, assume the answer is no until your privacy officer says yes.

EU AI Act (in force since 2024). AI built only for scientific research is exempt, but GDPR, your ethics committee and your university's rules still apply.

Terms of platforms and models. What you may collect, store, share, and train on.

The case for open models

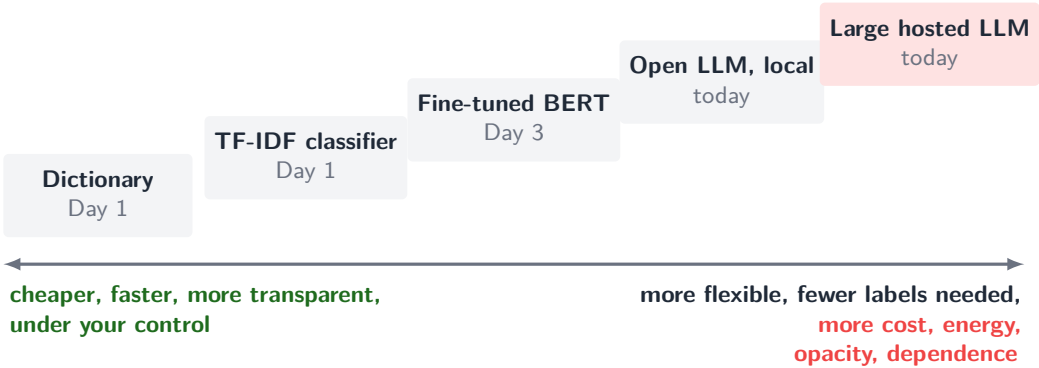
Open models are an **ethical imperative** for social scientists: you can see what you used, keep data private, and others can rerun it.

Spirling (2023), *Nature*

Using a proprietary model in academic research should require an **explicit justification**, like any other departure from good practice.

Palmer, Smith & Spirling (2024), *Nature Computational Science*

The Smallest Model That Does the Job



Climb a step only when the one below fails **on your validation data**. The TF-IDF baselines you've run all week are often harder to beat than they look. And distillation takes an LLM's labels back down to step 3.

Before You Pick a Model

About your data

Is it personal or sensitive?

Who wrote it, and did they agree to this use?

What do the platform's terms allow?

About the model

Can you download and archive it?

What does its license allow?

Is its training data documented?

How well does it work for your language, and for the groups you study?

About the cost

How many texts, how often?

What does each cost in time, money and energy?

Could a colleague with a smaller budget redo it?

If you can't answer these, you can't defend the choice in your methods section either.

Efficiency Is an Ethical Choice Too

Use less model

- Run a cheap baseline first, and stop if it's good enough.
- Try 1.5 billion parameters before 70 billion; use quantized versions locally.
- For a fixed task, prefer a task-specific model: fine-tune or distill.

Use the model less

- Short prompts, short answers: a label, not an essay. Label scoring over chain-of-thought.
- Test on a few hundred texts before the whole corpus.
- Save every answer; never send the same text twice.
- Feed the GPU batches of texts, not one at a time.

The *seconds per title* column in your notebook is an energy column too.

Good Practice for Models in Research

✔ Do

- Validate against human coding of your own data, with a test set you use once
- Report the model, version, route, prompt, settings and date
- Pre-register prompts and analyses where you can
- Prefer open weights you can archive; keep sensitive data local
- Check errors separately for the groups you study
- Disclose model use to readers, participants and your ethics board
- Pay human coders fairly

✘ Don't

- Treat model output as ground truth
- Send personal data to a commercial API without a legal basis
- Tune prompts on your test set
- Assume the model will behave the same next month
- Use a 70-billion-parameter model where TF-IDF does the job

Transparency ties it together: another researcher should be able to see what you did and, ideally, redo it.

Distillation: Pay the Big Model Once

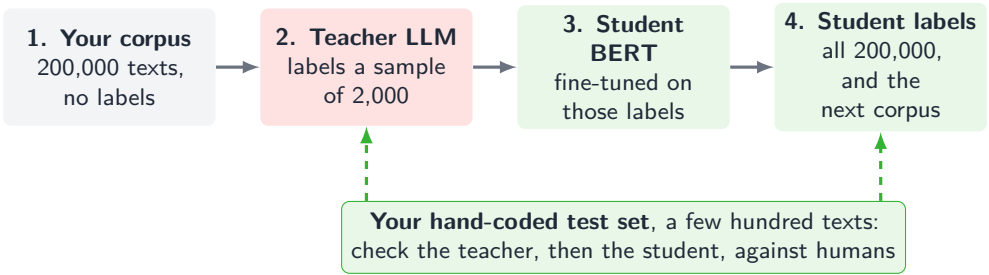
You don't have to choose between a powerful but expensive LLM and a small, cheap BERT. Use both, one after the other.

A **teacher** (a large LLM) labels a sample of your texts. A **student** (a small model like Day 3's BERT) is fine-tuned on those labels. Then the student labels everything: quickly, cheaply, on your own machine.

Hinton, Vinyals & Dean (2015) introduced the idea in machine learning, training small networks to imitate big ones. Social scientists mostly use the simplest version: the teacher's labels become the student's training data.

	Teacher LLM	**Student BERT**
Parameters	billions	about 100 million
Per text	seconds	milliseconds
Runs on	big GPU or API	laptop or Colab
Labels needed	none	the teacher's

The Distillation Workflow



You pay for the LLM once, on a sample. The student then labels this corpus, and the next one, for almost nothing.

What Distillation Fixes, and What It Doesn't

✔ It helps with

Cost and speed: milliseconds per text, no fees per token

Energy: a task-specific model, the short bar on the energy slide

Privacy at scale: the bulk of your data never goes to an API

Reproducibility: a small model you can archive and share

⚠ It doesn't fix

The student **inherits the teacher's errors and biases**, including the covert ones

Labels are only as good as the teacher: validate against humans, not the teacher

Rare categories need enough examples in the sample

Licenses: OpenAI's terms forbid using its output to build competing models; Llama's license asks for "Llama" in the name of shared models trained on its output. Apache 2.0 models like Qwen2.5 1.5B don't.

For many annotation projects, this is the sweet spot: an LLM's flexibility, a BERT's cost, and a model you control.

Your Turn: Pick the Pipeline

IN PAIRS Pick one scenario. Decide which model(s), where they run, how you validate, and name one ethical risk. Six minutes.

A

Two million Dutch social media posts on the nitrogen crisis. Stance on the farmers' protests?

B

300 transcribed patient interviews about their care, in Dutch. Themes and codes?

C

400 party manifestos from 25 countries, each in its own language. Positions on six policy scales?

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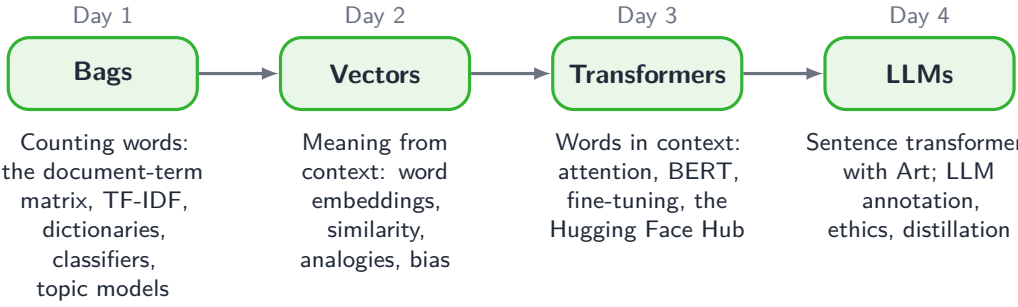
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One defensible answer each. **A:** an open LLM labels a sample locally; hand-code a few hundred posts to validate; distill into a Dutch BERT (e.g. RobBERT) for all two million. **B:** nothing leaves university servers: a local open model at most, human coding at the center. **C:** LLMs read whole documents, as in Benoit et al. (2026); check against expert surveys; repeat with more than one open model.

Four Days, One Arc



Each step added context and flexibility, and also cost, opacity and dependence. The newer tools don't replace the older ones: a dictionary, a TF-IDF classifier, a fine-tuned BERT and an LLM each have their place. The skill is choosing.

What to Take Home

1. **Start simple.** A baseline first; climb only when it fails.
2. **Validate everything** against human judgment, on texts you didn't tune on.
3. **Know what your measure captures**, and whose language it understands.
4. **Choose models deliberately**: data, labor, energy, access, and the company behind them.
5. **Make it redoable**: open models where you can, versions, code, and the full prompt.

Friday: optional online one-on-ones about your own project. **In the repo:** bonus notebooks on PDFs to text, transcription, scraping and APIs, anonymizing, topic models, text for causal questions, scaling up, and more.

Before You Go: Your Feedback

Please take **five minutes** now, before you leave.

Nine short questions, anonymous. Your answers help us improve the course, and help the graduate school decide whether to offer it again.

Scan the code, or use the link below it.



https://wur.az1.qualtrics.com/jfe/form/SV_eJQy8AbKOWUSBgy

Thank you!

From bags to vectors to transformers to LLMs.

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